



①

$q \rightarrow 10, 20, 5$

$w \rightarrow 70, 50, 30$

$k=2$ ✓

ans $\rightarrow 105.0000$

②

$q \rightarrow 3, 1, 10, 10, 1$

$w \rightarrow 4, 8, 2, 2, 7$

$k=3$

ans $\rightarrow 30.66669$

$\rightarrow$ minimise the total cost to hire k workers.

$$\begin{array}{ccc} w_1 & \rightarrow & q_x \\ & \searrow & \\ w_2 & \rightarrow & 2q_x \end{array} \quad \begin{array}{c} w_x \\ w_y \end{array}$$

$$\text{Pay} \geq w_y$$

$\hookrightarrow \underline{2w_x}$

We don't just need to pay min wage but also pay them proportionally.

→ if price is of major concern then among 2 workers we will pick the one with lesser cost.

w_1 w_2
 $mw_1 < mw_2$
(better choice) →

if the workers under comparison have min wage same → then pick one with better quality

Preference → low cost , high quality

$$\frac{mw}{9} \quad \} \quad \underline{\underline{Ratio}}$$

arrange worker in the ratio.

$$q \rightarrow 3, 1, 10, 10, 1$$

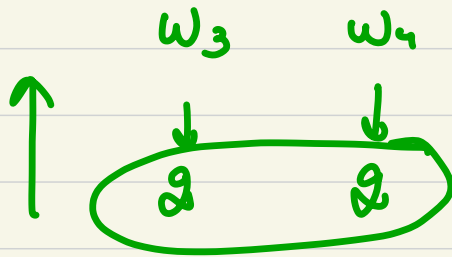
$$w \rightarrow 4, 8, 2, 2, 7$$

min wog

$$k = 3$$

$$\begin{array}{llll} \rightarrow & w_1 \rightarrow 4/3 & \rightarrow & 1033 \\ & w_2 \rightarrow 8/1 & \rightarrow & 8 \\ & w_3 \rightarrow 2/10 & \rightarrow & 0.2 \\ & w_4 \rightarrow 2/10 & \rightarrow & 0.2 \\ & w_5 \rightarrow 7/1 & \rightarrow & 7 \end{array}$$

$$\begin{array}{ll} q_2 \rightarrow 4 & q_4 = 2 \\ w_2 \rightarrow 10 & w_7 = 10 \end{array}$$



$$\begin{array}{c} w_1 \\ \downarrow \\ 4 \end{array}$$

$$w_5$$

$$w_2$$

$$\frac{w_1}{q_2} \rightarrow \frac{10}{4} \rightarrow 2.5$$

$$\frac{w_4}{q_4} \rightarrow \frac{10}{2} \rightarrow \underline{\underline{5}}$$

get K workers in min cost.

↳ heap

$\frac{7}{8}$

$$w_0, w_1, w_2$$

$$q \rightarrow 10, 20, 5$$

$$w_0, w_1, w_2$$

$$w \rightarrow 70, 50, 30$$

$$\underline{\underline{k=2}}$$

$$\rightarrow \frac{w}{q} \rightarrow 7$$

$$2.5 \quad 6$$

$$w_0$$

$$w_1$$

$$w_2$$



$$\underline{w_0}$$

$$\underline{\underline{120}} \quad \underline{\underline{50}} + \underline{\underline{30}}$$

$$\begin{array}{r} 30 + 70 \\ \hline 35 + 70 \\ \hline \underline{\underline{105}} \end{array}$$

pay either min wage or proportional last column is

high

$$w_{01} \rightarrow q_1 \rightarrow 50$$

$$M w_1 \rightarrow \underline{15}$$

$$w_{02} \rightarrow q_2 \rightarrow 10$$

$$M w_2 \rightarrow \underset{\curvearrowright}{\underline{1}} \rightarrow \underline{\underline{3}}$$



Say we have selected K workers

(Total
pay)



$$\left(\sum \text{quality of } K \text{ workers} \right) \times \left(\frac{w}{q} \right)$$

↳ higher

$$\omega_0$$

$$\omega_{01}$$

$$\omega_{02}$$

...

$$\left(q \times \frac{\omega}{g} \right)$$

$$p_0 \Rightarrow \left[q \times \frac{\omega}{g} \right]$$

==

$$q, x, i \rightarrow \underline{\underline{\min w_{of}}}$$

if there is no proportion
→

$$\left(r_i \Rightarrow \frac{w_i}{q_i} \right)$$

$$\left(q_0 \times r_0 + q_1 \times r_1 + q_2 \times r_2 - \dots \right)$$

$$\rightarrow \textcircled{w_k} \rightarrow r_k$$

$$\frac{w \uparrow}{q \uparrow} \textcircled{r_k} \rightarrow (r_0 \ r_1 \ r_2 \ \dots)$$

Most expensive water has light w/g ratio.

$$q \rightarrow 10, \quad \underline{20, 5}$$

$$w \rightarrow 70, \quad \underline{50, 30}$$

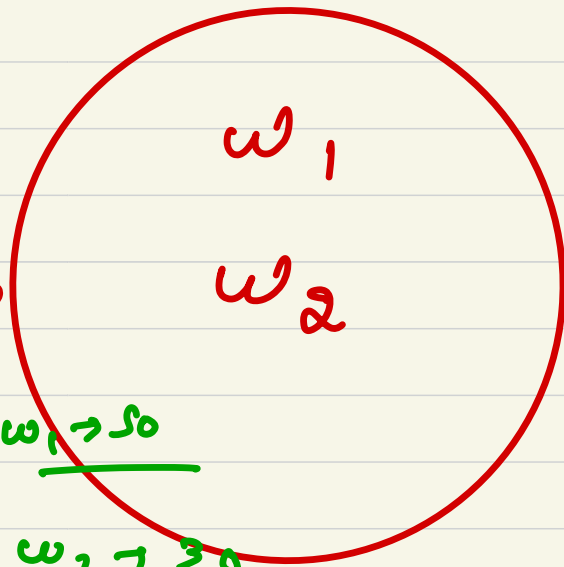
$$\underline{\underline{K=2}}$$

$$\rightarrow 7, \quad 2.5, \quad 6$$

$\frac{w}{q}$

$$\underline{w_1 \quad w_2} \quad w_0 \quad \leftarrow$$

$\rightarrow$ initially K workers



$$\underline{\underline{150}} \rightarrow \begin{array}{ll} w_1 \rightarrow 2.5 & q_1 \rightarrow 20 \quad \underline{w_1 \rightarrow 50} \\ w_2 \rightarrow 6 & q_2 \rightarrow 5 \quad \underline{w_2 \rightarrow 30} \end{array}$$

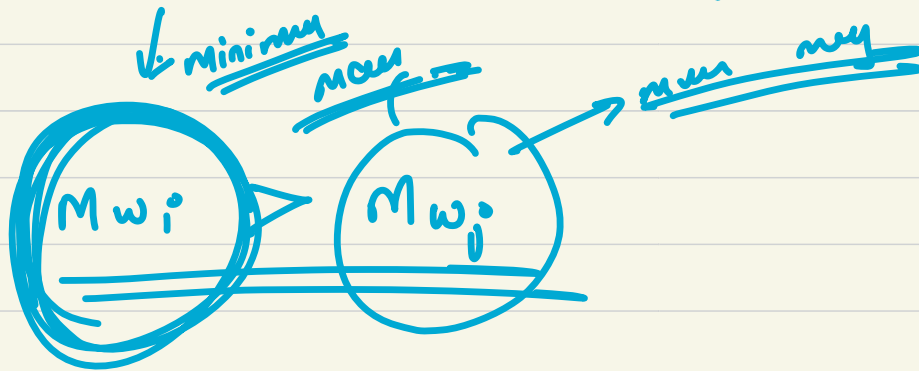
$$\rightarrow 30 + 50$$

$$\underline{r_j \leq r_i}$$

$$\longrightarrow \sigma_1 \leq \sigma_2$$

$$\sigma \leq i$$

$$\underline{tw_j} = \left(\overset{\nwarrow}{\sigma_j \times q_j}, \quad r_i \times q_j \right)$$

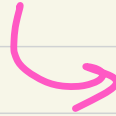
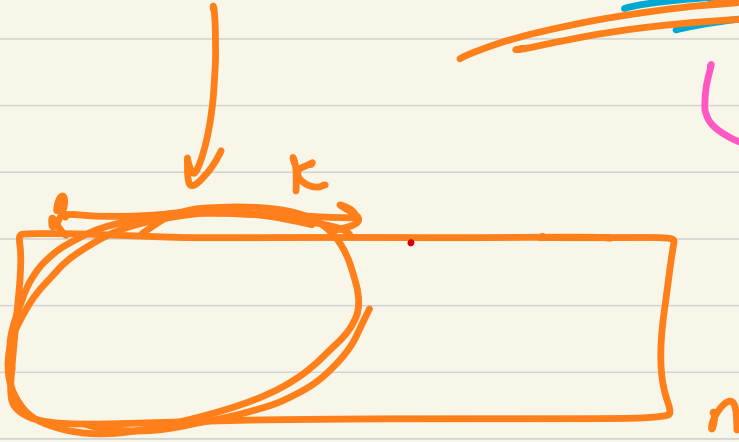


$$\underline{\underline{\gamma_i \geq \gamma_j}}$$

$$\gamma_n = \frac{w_n}{q_n}$$

$$(q_i \times \gamma_j + q_j \times \gamma_i)$$

Solved → based on ratio



Maxheap → quality

↓
max size → k

→ $\left((\text{quality}) \times \text{ratio of } k^{\text{th}} \text{ employee} \right)$

$$q \rightarrow \overset{\omega_0}{\underline{3}}, \overset{\omega_1}{4}, \overset{\omega_2}{1}, \overset{\omega_3}{6}, \overset{\omega_4}{9}, \overset{\omega_5}{8}, \overset{\omega_6}{2}$$

$$mw \rightarrow 10, 5, 3, 10, 20, 15, 12$$

$$\gamma \rightarrow \frac{10}{3}, \frac{5}{4}, 3, \frac{5}{3}, \frac{20}{9}, \frac{15}{8}, 6$$

$$\rightarrow \left(\underset{\omega_0}{3.3}, \underset{\omega_1}{1.25}, \underset{\omega_2}{\underline{3}}, \underset{\omega_3}{1.66}, \underset{\omega_4}{2.22}, \underset{\omega_5}{\underline{1.875}}, \underset{\omega_6}{6} \right)$$

$$\underline{\underline{K=3}}$$

$$\gamma_1 < \gamma_2$$

$$\underline{\underline{\frac{\omega_4}{q_4} < \frac{\omega_2}{q_2}}}$$

$$\rightarrow \overset{\omega_1}{\underline{\underline{\cancel{\omega_1}}}}, \overset{\omega_3}{\cancel{\omega_3}}, \overset{\omega_5}{\cancel{\omega_5}}, \overset{\omega_4}{\cancel{\omega_4}}, \overset{\omega_2}{\underline{\omega_2}}, \overset{\omega_0}{\omega_0}, \overset{\omega_6}{\omega_6}$$

$$\text{total quality} = \cancel{4+6} / \underline{\underline{18}}$$

$$\text{ans} = 18 \times 1.8 = \underline{\underline{33.75}}$$

$w_1 \rightarrow 7.2, w_3 \rightarrow 10.8, w_5 \rightarrow \underline{\underline{15}}$

$$4, \underline{\underline{1}}, 3$$

Max heap of quality

$$\text{Ans} \rightarrow 4 \times 3 + 6 \times 3 + 1 \times 3$$

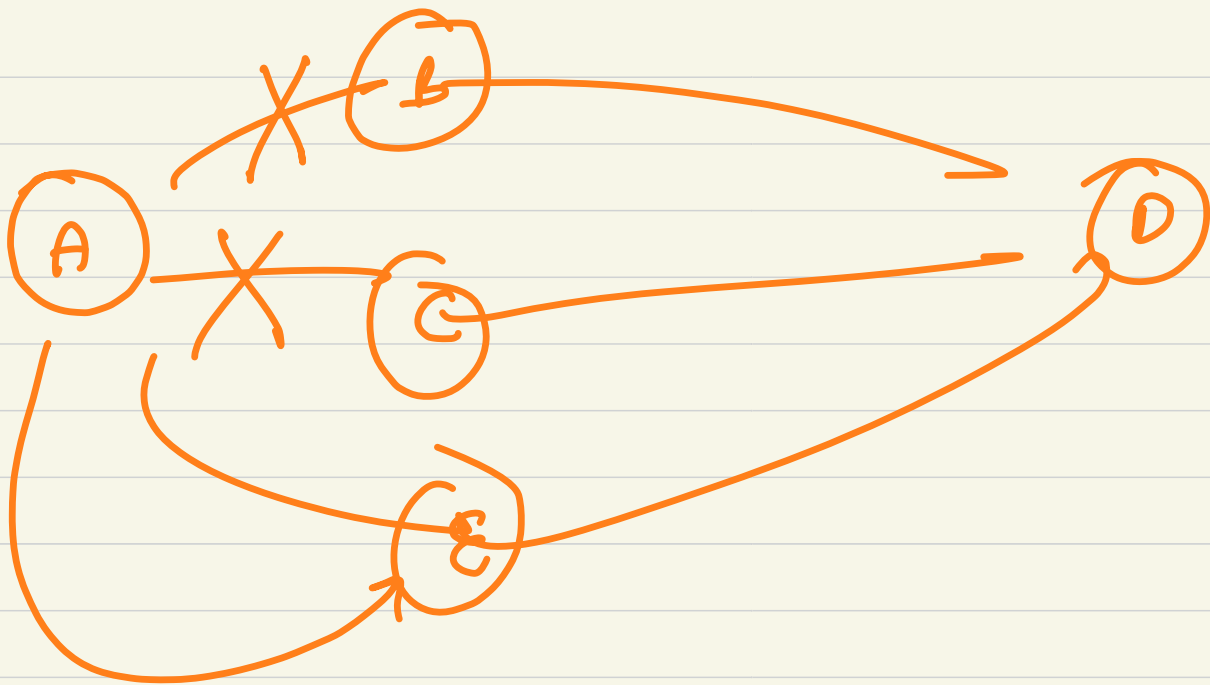
$$12 + 18 + 3 \rightarrow \underline{\underline{33}} \downarrow$$

$$\text{Ans} \rightarrow \underline{\underline{26.4}}$$

pick those workers with lowest quality &
pay $\frac{w}{q}$ ratio

(Σ quality) $\times \delta_n$

$\delta_n \rightarrow$ luck



$n \rightarrow 1$

3

9

$\downarrow 13$

3

$\downarrow 13$

1

-1

10

$\downarrow 12$

5

$\downarrow 7$

4

$\downarrow 12$

2

$\downarrow 17$

1

$\rightarrow 13$

$\rightarrow 12$

$\rightarrow -1$

4

S_1

S_2

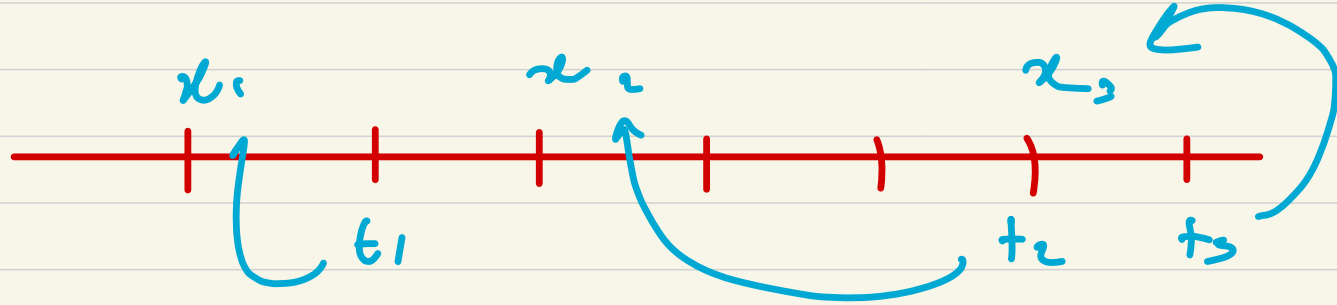
S_3

δt_1

δt_2

δt_3

$x \rightarrow \text{seat}$
 $t \rightarrow \text{schul}$



$$[a_1 \ a_2 \ a_3 \ \dots \ a_m]$$

median

$\frac{(m+n)}{2}$ th element

$$[b_1 \ b_2 \ b_3 \ \dots \ b_n]$$

$$a_{\frac{m}{2}} < b_{\frac{n}{2}}$$

a_1

$a_{\frac{m}{2}}$

a_m

b_1

$b_{\frac{n}{2}}$

b_n

$$(a_1 - a_{\frac{m}{2}} + b_1 - b_{\frac{n}{2}})$$

$$|| - ?$$

$$= \text{sr}$$

$$a_{\frac{m}{2}} > b_{\frac{n}{2}}$$

0

1 3

6

2

2

8

13

15

19

22

3

5

10

12

6

↑
1

3

6-2+1 → 4

5-1+1 → 3 (2)